MTH 201 Chapter 3: Infinite Series

3.7 The Ratio Test

In previous sections we have seen that in order for a series $\sum_{n=1}^{\infty} a_n$ to converge it must be the case that $\lim_{n\to\infty} a_n = 0$. However, we also know that series like the harmonic series satisfy the condition that $\lim_{n\to\infty} a_n = 0$ but does not converge. So not only do we need $\lim_{n\to\infty} a_n = 0$, but we need to have $a_n \to 0$ quickly enough. In this section we develop a test to help us with this distinction.

Theorem (Ratio Test): Let $\sum_{k=1}^{\infty} a_k$ be a series with nonzero terms and $\rho = \lim_{k\to\infty} \frac{|a_{k+1}|}{|a_k|}$.

1. If $\rho < 1$, the series $\sum_{k=1}^{\infty} a_k$ converges absolutely and therefore converges.
2. If $\rho > 1$ or $\rho = +\infty$, the series $\sum_{k=1}^{\infty} a_k$ diverges.
3. If $\rho = 1$, the test is inconclusive and no conclusion can be drawn.

Proof: The proof is in the textbook.

Note: As we are used to, this theorem still holds if the index starts at a value other than 1.

<u>Example 1:</u> Determine (with justification) whether the series $\sum_{k=1}^{\infty} \frac{1}{k!}$ converges or diverges.

Clearly the series contains all nonzero terms. So we use the Ratio test and compute

$\rho = \lim_{k\to\infty} \frac{|a_{k+1}|}{|a_k|} = \lim_{k\to\infty} \frac{\left(\frac{1}{(k+1)!}\right)}{\left(\frac{1}{k!}\right)} = \lim_{k\to\infty} \frac{1}{(k+1)!} \cdot \frac{k!}{1} = \lim_{k\to\infty} \frac{1}{k+1} = 0 < 1$. Therefore, by the ratio test, the series $\sum_{k=1}^{\infty} \frac{1}{k!}$ converges.

■

Example 2: Determine (with justification) whether the series $\sum_{k=1}^{\infty} \frac{k^k}{k!}$ converges or diverges.

Clearly the series contains all nonzero terms. So we use the Ratio test and compute

$$\rho = \lim_{k\to\infty} \frac{|a_{k+1}|}{|a_k|} = \lim_{k\to\infty} \frac{\left(\frac{(k+1)^{k+1}}{(k+1)!}\right)}{\left(\frac{k^k}{k!}\right)} = \lim_{k\to\infty} \frac{(k+1)^{k+1}}{(k+1)!} \cdot \frac{k!}{k^k} = \lim_{k\to\infty} \frac{(k+1)^k}{k!} \cdot \frac{k!}{k^k} = \lim_{k\to\infty} \frac{(k+1)^k}{k^k} = \lim_{k\to\infty} \left(\frac{k+1}{k}\right)^k$$

$$= \lim_{k\to\infty} \left(1 + \frac{1}{k}\right)^k = e > 1$$

Therefore, by the ratio test, the series $\sum_{k=1}^{\infty} \frac{k^k}{k!}$ diverges.

■

Example 3: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \left(\frac{(-1)^n 2^n}{n!}\right)$ is absolutely convergent, conditionally convergent, or divergent.

We first use the ratio test to test to test for absolute convergence/divergence.

We note that in this case $a_n = \frac{(-1)^n 2^n}{n!}$.

Now, we calculate that

$$\rho = \lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} = \lim_{n\to\infty} \frac{\left(\frac{2^{n+1}}{(n+1)!}\right)}{\left(\frac{2^n}{n!}\right)} = \lim_{n\to\infty} \frac{n!2^{n+1}}{(n+1)!2^n} = \lim_{n\to\infty} \frac{2}{n+1} = 0 < 1 .$$

Thus, this series is absolutely convergent by the ratio test (and therefore convergent!).

■

Example 4: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2-0.75}}$ is absolutely convergent, conditionally convergent, or divergent.

We first use the ratio test to test for absolute convergence/divergence. We note that in this case $a_n = \frac{(-1)^n}{\sqrt{n^2-0.75}}$. Now, we calculate that

$$\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} = \lim_{n\to\infty} \frac{\left(\frac{1}{\sqrt{(n+1)^2-0.75}}\right)}{\left(\frac{1}{\sqrt{n^2-0.75}}\right)} = \lim_{n\to\infty} \sqrt{\frac{n^2-0.75}{(n+1)^2-0.75}} = \lim_{n\to\infty} \sqrt{\frac{n^2-0.75}{n^2+2n+.25} \cdot \frac{1/n^2}{1/n^2}} = \sqrt{\lim_{n\to\infty} \frac{1-.75/n^2}{1+2/n+.25/n^2}} = 1 \ .$$

Thus, the Ratio Test is Inconclusive. So, the Ratio Test has gotten us nowhere.

Suppose now we try the Alternating Series Test. We will work with $\frac{1}{\sqrt{n^2-0.75}}$. Let $b_n = \frac{1}{\sqrt{n^2-0.75}}$ We calculate that $\lim_{n\to\infty} b_n = 0$. Also since $f(x) = \sqrt{x}$ is increasing on $(0,\infty)$, we know that for any natural number n

$$0 < \sqrt{n^2-0.75} < \sqrt{(n+1)^2-0.75} \Rightarrow \frac{1}{\sqrt{(n+1)^2-0.75}} < \frac{1}{\sqrt{n^2-0.75}} \Rightarrow b_{n+1} < b_n.$$

So, we know that the series is convergent by the Alternating Series Test.

So, the series is absolutely convergent or conditionally convergent. To determine which of these it is we consider $\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{\sqrt{n^2-0.75}} \right| = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2-0.75}}$. We will compare this series to the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$. We note that $n^2-.75 < n^2 \Rightarrow \sqrt{n^2-.75} < n \Rightarrow \frac{1}{n} < \frac{1}{\sqrt{n^2-.75}}$ where n is any natural number. Since the harmonic series is divergent, we know by the Comparison Test that $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2-0.75}}$ is divergent. So the series is not absolutely convergent. Therefore, we conclude that the series is conditionally convergent.

■